which is certainly true, since `straight_line(v,u)≤road_distance(v,u)`, and straight-line distance, being an Euclidean distance, certainly abide by triangle inequality[14].

While this condition always holds, the value of the heuristic for a vertex u can be larger than the value assigned to its parent v; for example, in figure 14.22, consider vertices v_E and v_D: the straight-line distance between vertex v_D and the goal is larger than its parents', v_E's; this happens because not all vertices are connected by an edge (or rather, two directional edges), and so for example from v_E (which is the closest vertex to the goal) you can reach v_G only with a detour, by first going through v_D.

It can be proved that if, instead, the graph was fully connected, we would also visit vertices according to their straight-line distance from the goal: in other words, the value of the heuristic would monotonically decrease during the traversal.

Then again, most useful graphs are not fully connected, and luckily we don't need this strict condition for A* to be able to find the optimal solution, but we can make do with a consistent and admissible heuristic.

These properties can guarantee us that the best solution will be found, but should we also aim to get the most accurate estimate possible? There might be many admissible and consistent heuristics, does the algorithm find the best rout faster if we choose one with a more precise estimate?